THOMAS LENZ

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OBJECTIVE

To leverage human factors knowledge of human abilities, limitations, and needs to build human-centered technology and environments. To learn, through human subjects experiments, how to build more responsible automation. To teach, mentor, and disseminate human factors knowledge and best practices.

RESEARCH INTERESTS

- Evaluation of human-vehicle interfaces using physiological and behavioral signals from simulator experiments
- Management of operator workload and situation awareness in autonomous vehicle teleoperation queues
- Design and evaluation of chatbots that encourage appropriate trust and acceptance in users

EDUCATION

Ph.D., Industrial Engineering and Human-Computer Interaction, Iowa State University *Spring 2029*

M.S., Industrial Engineering, Iowa State University *Spring 2027*

B.I.Sy.E., Industrial Engineering, University of Minnesota-Twin Cities *Spring 2024*
Minor in Computer Science

EMPLOYMENT HISTORY

Iowa State University, Ames, IA
Harold and Shirley Reihman Graduate Scholar Research Assistant - HIA Lab *Fall 2024-Present*

- Design and evaluate human-vehicle interfaces in simulated automated driving environments to assess operator workload, trust, and situational awareness
- Mentor undergraduate researchers on experiment preparation, simulator development, and data analysis
- Collecting data from 32 participants on a UE4 driving simulator to assess display compellingness during partially-automated driving
- Developing an open-source CARLA- and Python API-based simulator for assessing situation awareness during remote assistance of autonomous vehicles

TA - Applied Ergonomics and Work Design *Spring 2026, Spring 2027*

- Developed cognitive ergonomics curriculum
- Taught two lectures on cognitive ergonomics
- Held well-attended office hours

- Assisted with class activities and labs

Project Mentor - NSF Research Experience for Undergraduates *Summer 2025*

- Mentored a group of three undergraduate students in a summerlong research project
- Developed a CARLA-based simulator to test the effects of queue design on automated vehicle remote operator performance
- Helped students formulate a research question, develop an experiment plan, build a simulator, and test experiment subjects
- Resulted in production of a prototype teleoperation simulator, a pilot study, a draft paper, and a poster presentation

University of Minnesota, Minneapolis, MN
Undergraduate Research Assistant - HumanFIRST Lab *Fall 2023-Fall 2024*

- Collected and analyzed data on pedestrian crossing behavior and driver yielding behavior
- Evaluated efficacy of environmental interventions on pedestrian safety

Jr. Engineer Trainee - St. Anthony Falls Laboratory *Fall 2021-Fall 2023*

- Assisted professors and graduate students in experimental setup
- Fabricated metal, plastic, and wood apparatuses

MVE, New Prague, MN
Product Engineering Intern *Summer 2023*

- Developed testing protocol to assess dewar durability
- Designed experiments to minimize cost of testing while maximizing statistical power
- Evaluated usability of dewars

PUBLICATIONS

3. **Lenz, T.**, EJ Bahng, S Park, J Liu, & B Weng (2026). A Survey of Future Teachers Suggests Cautious Openness to VR and Humanoid Robot Use. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. https://doi.org/10.1177/10711813261475235
2. **Lenz, T.**, Brackey, R., & Liu, J. (2025). The effects of citations and confirmation bias on trust in chatbots. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. https://doi.org/10.1177/10711813251357884
1. Craig, C., Mabry, M., Drahos, B., Kessler, W., **Lenz, T.**, Olander, K., Morris, N. (2024). An assessment of two pedestrian safety inventories. *Advances in Human Factors of Transportation*. AHFE International Conference. http://doi.org/10.54941/ahfe1005242

OTHER SCHOLARLY ACTIVITIES

Proposals

"Collaborative Research: SCH: VR-Enhanced Digital Twin for Adaptive Nurse Fatigue Management," *NSF Smart and Connected Health Program* [coauthor, under review]. Spring 2026.

Conference Presentations

6. "A Survey of Future Teachers Suggests Cautious Openness to VR and Humanoid Robot Use," *ASPIRE: Human Factors and Ergonomics Society Annual Meeting* [poster presentation]. Reno, NV, Fall 2026.
5. "A Systematic Review of Human Factors Solutions to Improve Autonomous Vehicle Remote Assistant Situation Awareness," *IISE Annual Conference* [lecture]. Arlington, TX, Spring 2026.
4. "A Systematic Review of Human Factors Solutions to Improve Autonomous Vehicle Remote Assistant Situation Awareness," *Iowa State University IMSE Spring Poster Symposium* [poster presentation]. Ames, IA, Spring 2026.
3. "The Effects of Citations and Confirmation Bias on Trust in Chatbots," *ASPIRE: Human Factors and Ergonomics Society Annual Meeting* [poster presentation]. Chicago, IL, Fall 2025.
2. "The Effects of Citations and Confirmation Bias on Trust in Chatbots," *Iowa State University IMSE Spring Poster Symposium* [poster presentation]. Ames, IA, Spring 2025.
1. "Effect of Personalized Driving Algorithms on Trust and Workload," *HFES Midwest Regional Meeting* [lightning talk]. West Lafayette, IN, Spring 2025.

Guest Lectures

4. "Cognitive Ergonomics Sampler," *IE 2710 Applied Ergonomics and Work Design*. Iowa State University, Spring 2026.
3. "About HIA Lab," *IMSE Departmental Recruitment Events and Open Houses* [about one presentation per month]. Iowa State University, Spring 2025 - Present.
2. "My Research Journey," *HCI 5790X Methods for Interdisciplinary Research*. Iowa State University, Fall 2025.
1. "HFES Student Chapter Trivia," *HFES at ISU*. Iowa State University, Fall 2025.

Professional Society Leadership

- **President**, HFES Student Chapter at Iowa State University *Spring 2026-Present*
- **Senator**, ISU Graduate and Professional Student Senate *Fall 2025-Present*
- **Vice President**, HFES Student Chapter at Iowa State University *Fall 2025-Spring 2026*
- Member, Institute of Industrial and Systems Engineers (IISE) *Spring 2026-Present*
- Member, Human Factors and Ergonomics Society (HFES) *Fall 2024-Present*
- Member, Human Computer Interaction (HCI) Student Group *Fall 2025-Present*

SKILLS

Methods: User evaluation, survey design, statistical modeling, design of experiments, human subjects research, machine learning

Programming and Software: Python, R, Excel, MATLAB, Simio, AMPL, Qualtrics, Generative AI

Simulation Development: Unreal Engine, Unity, CARLA, VR, eye tracking, physiological signal collection, CAD

STUDENT MENTORING

Undergraduate Students Mentored

Name	Program	Graduation Year
Madeleine Korver	NSF REU	2027
Johann Chen	NSF REU	2027
Rindirisia Wangira	Undergraduate Research Assistant	2027
Nathan Phillips	NSF REU	2028
Garrett Galvin	Undergraduate Research Assistant	2028
Allison Vaughn	Undergraduate Research Assistant	2028
Adriana Soto	Undergraduate Research Assistant	2028
Navin Nagavel	Undergraduate Research Assistant	2029
Victoria Tran	Undergraduate Research Assistant	2030